

Author: But the reason is not confined only to the infinitely increasing frequency of oscillations of the graph. Another reason is the constancy of the amplitude of oscillations. Let us "slightly correct" our function by multiplying $\sin \frac{1}{x}$ by x . The graph of the function $y = x \sin \frac{1}{x}$ is shown in Fig. 26. Do you think that $b=0$ is the limit of this function at $x=0$?

Reader: I am at a loss.

Author: I will answer this question myself. Yes, it is. The proof is within your reach if you use the definition of the limit of function. You are welcome.

Reader: We select an arbitrary $\varepsilon > 0$. We should find $\delta > 0$ such that $|x \sin \frac{1}{x} - 0| < \varepsilon$ for all x (excluding $x=0$) satisfying the condition $|x - 0| < \delta$. It seems to me that δ we look for is $\delta = \varepsilon$.

Author: You are quite right. Because if $|x| < \delta = \varepsilon$, it becomes evident that $|x \sin \frac{1}{x}| = |x| |\sin \frac{1}{x}| < \varepsilon$ (since $|\sin \frac{1}{x}| \leq 1$).

↳ beautiful proof

Reader: Really, the existence of the limit is proved without considerable difficulties.

Author: But, certainly, not always. Consider, for example, a well-known function $y = \sqrt{x}$ and prove (using the definition of the limit of function) that $b=1$ is the limit of the function at point $x=1$.

To begin with, consider the following inequality: $|\sqrt{x} - 1| < \varepsilon$. Try to find a function $g(\varepsilon)$ such that $|x - 1| < g(\varepsilon)$ for any x satisfying the condition $|\sqrt{x} - 1| < \varepsilon$.

Reader: I understand that $g(\varepsilon)$ is actually the desired δ corresponding to an arbitrary ε .

Author: Yes, of course. We begin with some transformations. We shall proceed from the inequality: $|\sqrt{x} - 1| < \varepsilon$ (3) which can be rewritten in the form: $(1 - \varepsilon) < \sqrt{x} < (1 + \varepsilon)$